

Empowering Women, Securing Food: How Patriarchal Norms and Marital Status Shape Women’s Access to Resources and Food Security in Syria

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Introduction

This research explores how socio-demographic characteristics associated with women’s empowerment—specifically age and marital status—influence household food insecurity among female-headed households (FHHs) in Syria, and how these patterns vary across governorates. In fragile and conflict-affected settings, where social, political and institutional instability intersects with entrenched gender inequalities, household food security is particularly precarious (Doocy et al., 2017). FHHs are often among the most vulnerable in these contexts, although their specific experiences remain underexplored, especially in countries where traditional gender roles persist despite significant social change.

Following the Syrian conflict that began in 2011, widespread male displacement and fatalities forced many women into provider and decision-making roles. However, this shift in responsibility did not coincide with structural or legal reforms. Women remain largely excluded from property ownership, financial services and protective legislation, demonstrating that headship does not automatically equate to meaningful empowerment (UNDP, 2018; SCPR, 2013).

Grounded in Kabeer’s (1999) framework, which defines empowerment through access to resources, agency in decision-making, and the ability to achieve valued goals, this analysis considers how marital status and age affect women’s capacity to navigate patriarchal constraints. Marital status functions as a structural marker: in Syrian society, women often access legitimacy and support through male kin. Married women may benefit from these ties, while widows—despite being de facto heads of household—frequently face marginalization and diminished institutional access. Age, however, may mitigate some of these challenges.

Older women often hold greater social credibility, allowing them to exercise informal authority and negotiate access in ways younger widows cannot. In this sense, age can serve as a compensatory resource, softening the disempowerment associated with widowhood.

The following hypotheses guide the analysis:

- **H1:** Married female household heads report lower food insecurity than widows, due to enhanced legitimacy and access to resources.
- **H2:** Among widows, older women experience lower food insecurity than younger women, as age confers more social credibility.
- **H3:** Baseline levels of food insecurity vary by governorate, suggesting that contextual factors may shape women’s experiences of hardship.

Governorates are modeled as varying intercepts, capturing unobserved regional effects without asserting direct causal interpretations. In a rural, agrarian-based economy like Syria’s, they serve as proxies for broader opportunity structures shaped by geography, gender norms and conflict exposure. In conclusion, the analysis employs a non-centered Bayesian hierarchical model with partial pooling to account for the cross-classified structure of the data and the small number of observations per group. The outcome variable is a log-transformed version of the Reduced Coping Strategy Index (RCSI), modeled as continuous.

Data and Variables

This study draws on baseline data collected in Syria in 2023, before Ramadan, comprising 466 female-headed households from five governorates: Hama, Deir-ez-Zor, Aleppo, Homs, and Rural Damascus. All variables are measured at the household level, with sociodemographic information specifically referring to the household head.

For marital status, the analysis focuses on married and widowed women, as divorced and single women were excluded due to limited sample size. Age was recoded into four life-course categories-Young (18–29), Early Adult (30–44), Mid Adult (45–59), and Senior (60+)-which serve as group-level predictors in the hierarchical model (Table 1).

I. Outcome Variable: Reduced Coping Strategy Index (RCSI)

RCSI is a continuous measure of food insecurity (range: 0–42) based on frequency and severity of five coping behaviors during food shortages (Maxwell et al., 2013):

$$\text{RCSI} = \sum_{i=1}^5 (\text{frequency}_i \times \text{severity weight}_i) \quad (1)$$

Due to its right-skewed distribution (min = 0.47, median = 9.8, mean = 10.96, max = 35), the outcome was log-transformed using $\log(1 + \text{RCSI})$ to stabilize variance and improve distributional symmetry. The log-transformed RCSI is modeled as a continuous outcome.

Bayesian Model Specification

The goal of this model is to assess how household food insecurity varies across demographic and regional characteristics among female-headed households in Syria. The analysis accounts for variation by age group, marital status, governorate and the interaction between age and marital status.

The structure of the dataset supports a Bayesian hierarchical approach with partially overlapping (cross-classified) groupings. Because each demographic factor appears at multiple levels of the others, and interactions span across governorates, a non-nested, cross-classified model is most appropriate.

I. Modeling Approach:

This analysis employs a Bayesian hierarchical model with non-centered parameterization and partial pooling. Non-centered parameterization improves computational efficiency and convergence, especially in unbalanced groupings. Partial pooling stabilizes estimates in sparsely populated groups by borrowing strength from the overall distribution, as advocated by (Gelman et al., 2013).

The model includes weakly informative priors to regularize inference and reduce overfitting while maintaining flexibility to reflect real-world variability. The priors are not based on the observed data but aim to generate plausible values consistent with the research context (Gelman et al., 2017).

II. Modeling Structure:

Let y_n denote the log-transformed RCSI score for household n . The model is:

$$\begin{aligned} y_n &\sim \text{Normal}(\mu_n, \sigma) \\ \mu_n &= \alpha_{a[n]} + \gamma_{m[n]} + \delta_{g[n]} + \eta_{am[n]} \end{aligned}$$

Where:

- $\alpha_{a[n]}$ is the age group intercept,
- $\gamma_{m[n]}$ is the marital status effect,
- $\delta_{g[n]}$ is the governorate effect,
- $\eta_{am[n]}$ is the age $\times$ marital status interaction effect.

III. Priors and Justification:

We use a non-centered parameterization for all group-level effects:

$$\begin{aligned}\alpha_a &= \mu_\alpha + \sigma_\alpha \cdot \alpha_a^{\text{raw}}, & \alpha_a^{\text{raw}} &\sim \mathcal{N}(0, 1) \\ \gamma_m &= \sigma_\gamma \cdot \gamma_m^{\text{raw}}, & \gamma_m^{\text{raw}} &\sim \mathcal{N}(0, 1) \\ \delta_g &= \sigma_\delta \cdot \delta_g^{\text{raw}}, & \delta_g^{\text{raw}} &\sim \mathcal{N}(0, 1) \\ \eta_{am} &= \sigma_\eta \cdot \eta_{am}^{\text{raw}}, & \eta_{am}^{\text{raw}} &\sim \mathcal{N}(0, 1)\end{aligned}$$

Hyperpriors:

$$\begin{aligned}\mu_\alpha &\sim \text{Normal}(2.4, 0.3) \\ \sigma_\alpha, \sigma_\gamma, \sigma_\delta, \sigma_\eta, \sigma &\sim \text{Normal}^+(0, 0.2)\end{aligned}$$

Justification:

- μ_α : Centered on 2.4 to reflect the log-scale of the expected RCSI ($\exp(2.4) \approx 11$), based on contextual knowledge of food insecurity in Syria (Doocy et al., 2017).
- **Group-level SDs**: Weakly informative truncated normals, encouraging moderate regularization without overshrinking.
- **Residual SD (σ)**: Same prior as group-level SDs, avoiding overdispersed predictions while capturing household-level variability.

Model Quality Checks

To assess model performance and convergence, a series of diagnostics were conducted following the posterior sampling.

Convergence Diagnostics. The model displayed strong convergence across all chains. The $\hat{R}$ statistic was ≤ 1.01 for all monitored parameters, indicating that all chains mixed

well. Additionally, both bulk and tail effective sample sizes (ESS) were high, with values respectively around 2000 and 4500, suggesting sufficient exploration of the posterior distribution and robustness of estimates (See Appendix Fig. 3).

Prior Predictive Checks. Prior predictive simulations were conducted to ensure that the priors generated plausible values for RCSI in the Syrian context. Using weakly informative priors—centered at $\mu_\alpha \sim \text{Normal}(2.4, 0.3)$ and standard deviations drawn from $\text{Normal}^+(0, 0.2)$ —the prior predictive distribution of RCSI was concentrated between 5 and 25, with a central tendency around 11. This reflects reasonable prior uncertainty based on existing literature and empirical expectations (See Appendix Fig. 4).

Posterior Predictive Checks. Posterior predictive simulations were performed and visually compared to the observed log-RCSI distribution. The simulated values closely matched the observed distribution, with no significant over- or under-estimation across the support of the data. This indicates that the model captures the empirical structure of the outcome while remaining regularized by prior beliefs (See Appendix Fig. 5).

Results

Figure 1 displays the estimated varying intercepts for age group, marital status, and governorate, derived from the hierarchical Bayesian model. These reflect the relative contribution of each factor to predicted household food insecurity (log-transformed RCSI).

The effect of marital status indicates that widowed women experience slightly higher food insecurity than married women. However, the 89% credible intervals overlap zero, suggesting limited statistical certainty. While this provides only partial support for Hypothesis 1 (H1), the direction of the effect is consistent with theoretical expectations.

Age-related effects appear modest and uncertain. Although younger women tend to show slightly lower predicted food insecurity than older groups, the intervals across age levels substantially overlap. This implies little reliable evidence of a systematic age gradient, particularly among widows, offering no support for Hypothesis 2 (H2).

Governorate-level estimates reveal more pronounced differences, with regions such as Rural Damascus and Al-Hasakeh deviating from the overall mean. These results suggest that context-specific factors shape baseline food insecurity levels, aligning with Hypothesis 3 (H3).

Figure 1: Estimated Effects by Group: Age, Marital Status, and Governorate

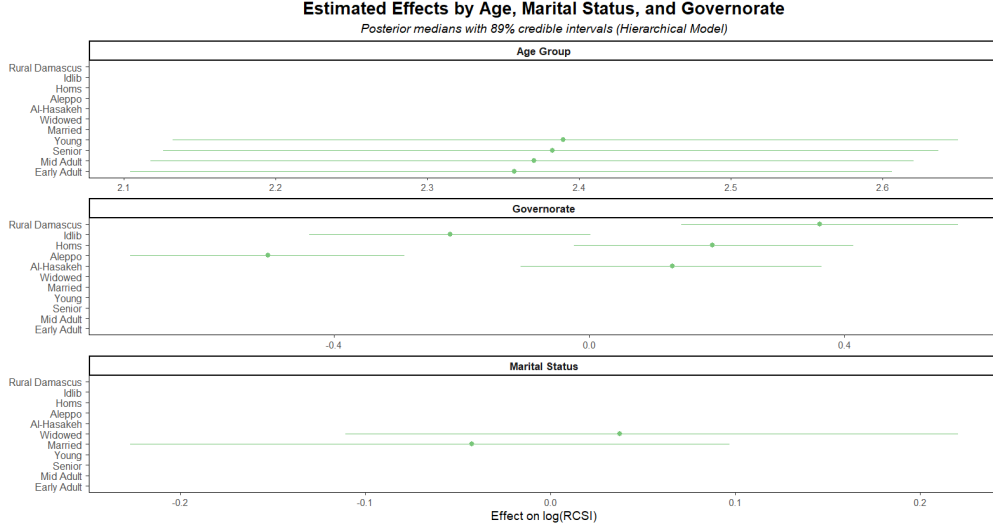


Figure 2 further explores these dynamics by displaying posterior predictions across all age-by-marital-status combinations, separately for each governorate. Across all regions, widowed women consistently exhibit higher predicted food insecurity than their married counterparts of the same age group. Although the credible intervals frequently overlap, the directional consistency reinforces partial support for H1.

In contrast, the data provide no consistent evidence that older widows experience lower food insecurity than their younger counterparts in any governorate, challenging the assumption that age confers greater social legitimacy and protection from vulnerability.

Finally, the extent and structure of age-by-marital differences clearly vary across governorates. The gap between groups is wider in some regions (e.g., Al-Hasakeh and Idlib) and narrower in others (e.g., Homs), indicating that demographic effects are mediated by local social and institutional contexts, the idea that regional context shapes baseline food insecurity levels, as proposed in H3..

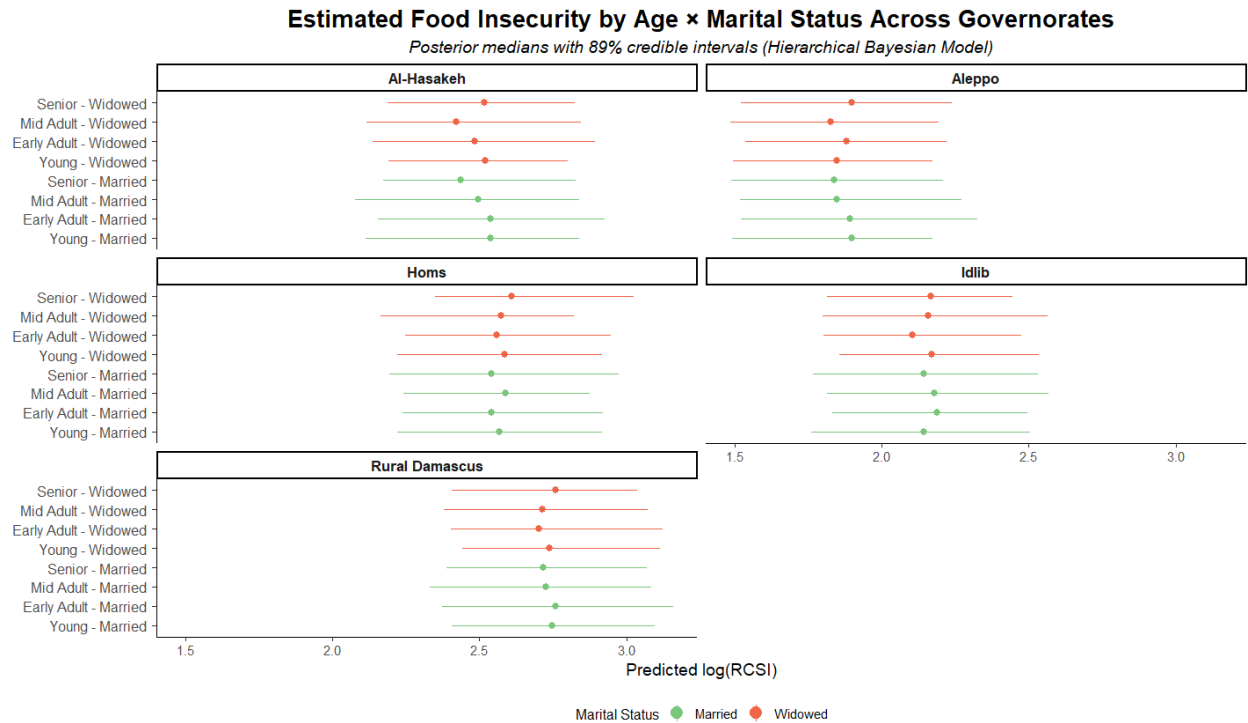


Figure 2: Estimated Food Insecurity by Age × Marital Status Across Governorates. Posterior medians with 89% credible intervals (Hierarchical Bayesian Model).

Discussion

While this study offers important insights into how age, marital status, and regional context shape food insecurity among female-headed households in Syria, several limitations warrant consideration.

First, the sample size within specific subgroups—particularly certain age-by-marital combinations within individual governorates—may be insufficient to yield precise posterior estimates. The use of a non-centered hierarchical model with partial pooling appropriately mitigates overfitting in sparse cells, but also contributes to conservative estimates by shrinking group-level effects toward the global mean. As a result, even patterns that are directionally consistent may not reach high posterior credibility.

Second, although the hierarchical structure allows for varying intercepts across key demographic and geographic dimensions, the model does not include other potentially important covariates such as household income, education, employment, or number of dependents. These factors may confound or mediate the relationships under study and could mask heterogeneity within marital or age groups.

Third, the outcome variable—log-transformed RCSI—is a composite food insecurity mea-

sure that captures short-term coping behavior rather than chronic deprivation. It may therefore understate structural inequalities that play out over longer timeframes or through different forms of vulnerability.

Fourth, the analysis focuses exclusively on baseline data from 2023, limiting temporal generalizability. Longitudinal data would be necessary to assess how demographic effects on food insecurity evolve over time, particularly in response to policy interventions or shifts in local conditions.

Finally, the model assumes additive contributions from each hierarchical level. While the age-by-marital interaction term captures some complexity, it does not account for potential three-way interactions involving governorates. Future work could explore these interactions explicitly, particularly given the strong contextual differences observed.

Despite these limitations, the model yields valuable insights by combining theory-driven groupings with Bayesian regularization. The results suggest that demographic vulnerabilities are not uniform across space, and underscore the importance of accounting for both social and geographic structure when designing humanitarian responses.

Table 1: Parameter Definitions and Prior Distributions (Final Model)

Parameter	Name	Prior	Scale	Interpretation
Age Group Effects (age_id)				
μ_α	Global log-RCSI mean	Normal(2.4, 0.3)	Log scale	Prior expected RCSI $\approx \exp(2.4) \approx 11$
σ_α	Age SD	Normal(0, 0.2)	Positive	Variation across age groups
α_{raw}	Age raw effect	Normal(0, 1)	Standardized	Latent z-scores for age
α	Age effect	Derived	Composite	$\mu_\alpha + \sigma_\alpha \cdot \alpha_{\text{raw}}$
Marital Status Effects (marital_id)				
σ_γ	Marital SD	Normal(0, 0.2)	Positive	Variation across marital groups
γ_{raw}	Marital raw effect	Normal(0, 1)	Standardized	Latent z-scores for marital status
γ	Marital effect	Derived	Composite	$\sigma_\gamma \cdot \gamma_{\text{raw}}$
Governorate Effects (gov_id)				
σ_δ	Governorate SD	Normal(0, 0.2)	Positive	Variation across governorates
δ_{raw}	Governorate raw effect	Normal(0, 1)	Standardized	Latent z-scores for governorates
δ	Governorate effect	Derived	Composite	$\sigma_\delta \cdot \delta_{\text{raw}}$
Age $\times$ Marital Interaction (age_marital_id)				
σ_η	Interaction SD	Normal(0, 0.2)	Positive	Variation for interaction effects
η_{raw}	Interaction raw effect	Normal(0, 1)	Standardized	Latent z-scores for interaction
η	Interaction effect	Derived	Composite	$\sigma_\eta \cdot \eta_{\text{raw}}$

Figure 3: Convergence Diagnostics

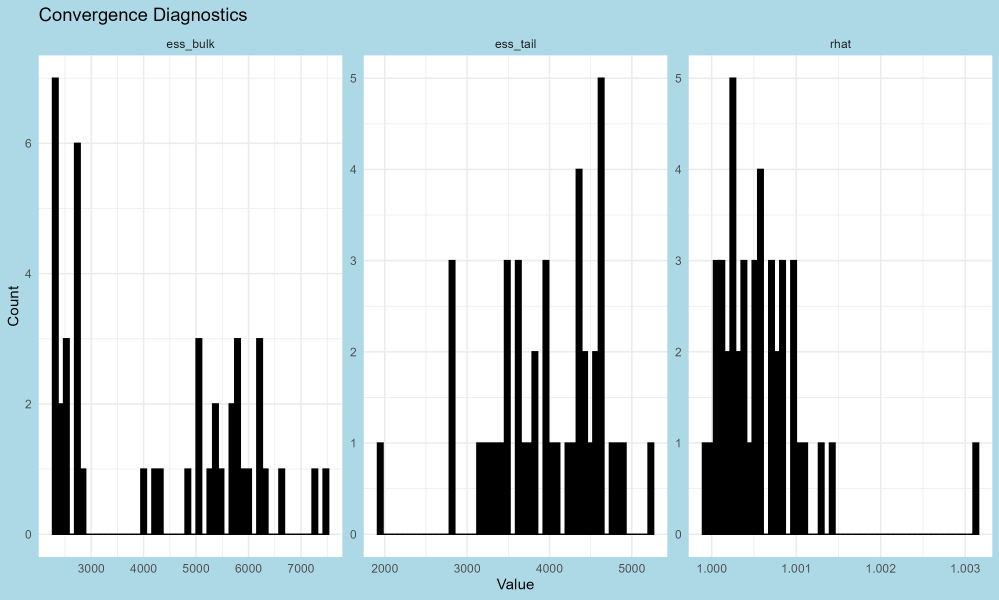


Figure 4: Prior Predictive Check

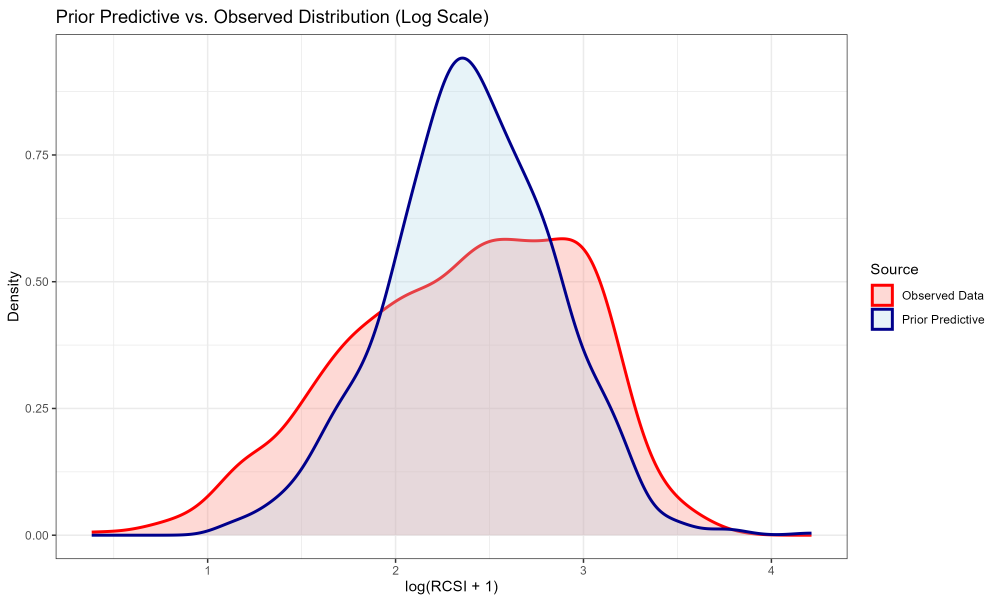
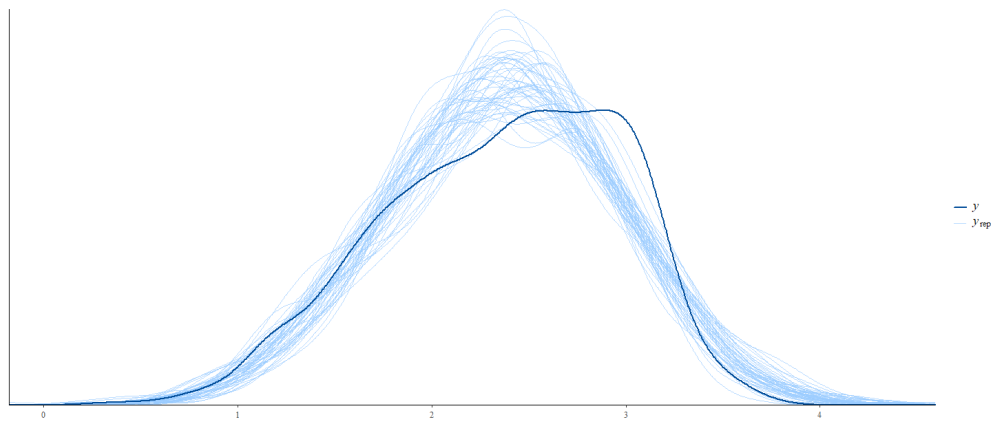


Figure 5: Posterior Predictive Check



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